$|\psi\rangle$

Day 1

Intro to Quantum Computing

KAUST Academy + KGSP

Day 1 · Quantum Déjà Vu

I’ve been in this place before

Imagine it's 1945

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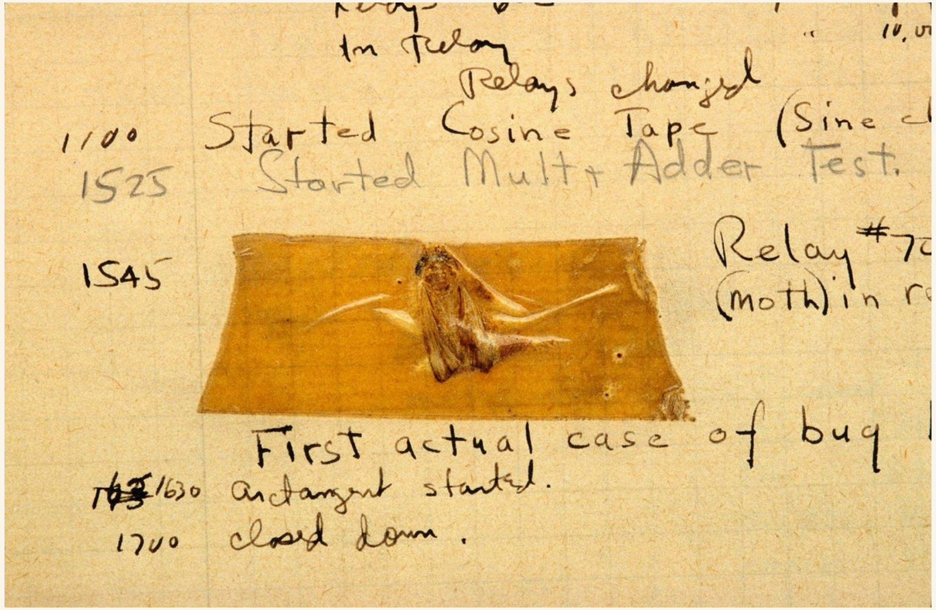
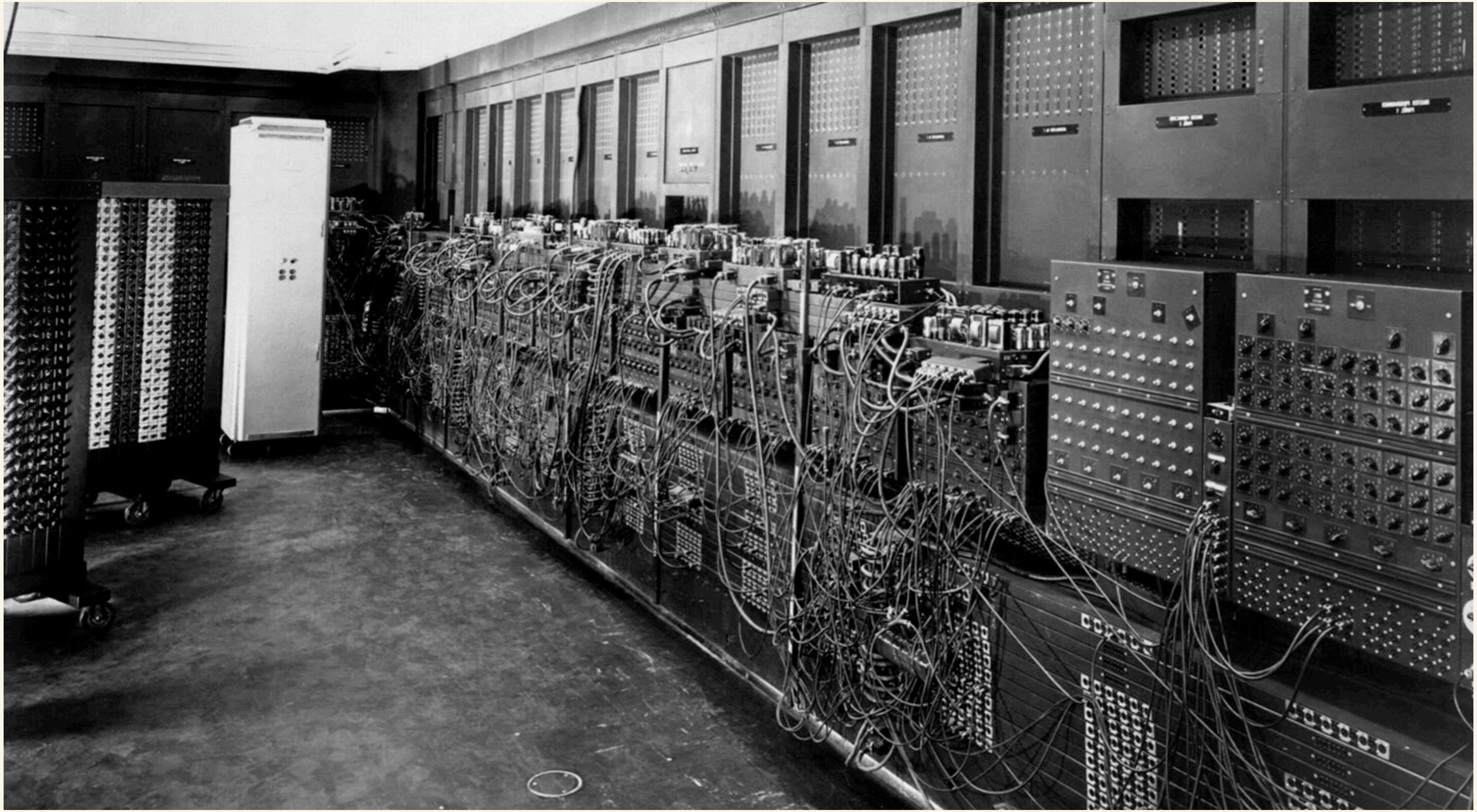


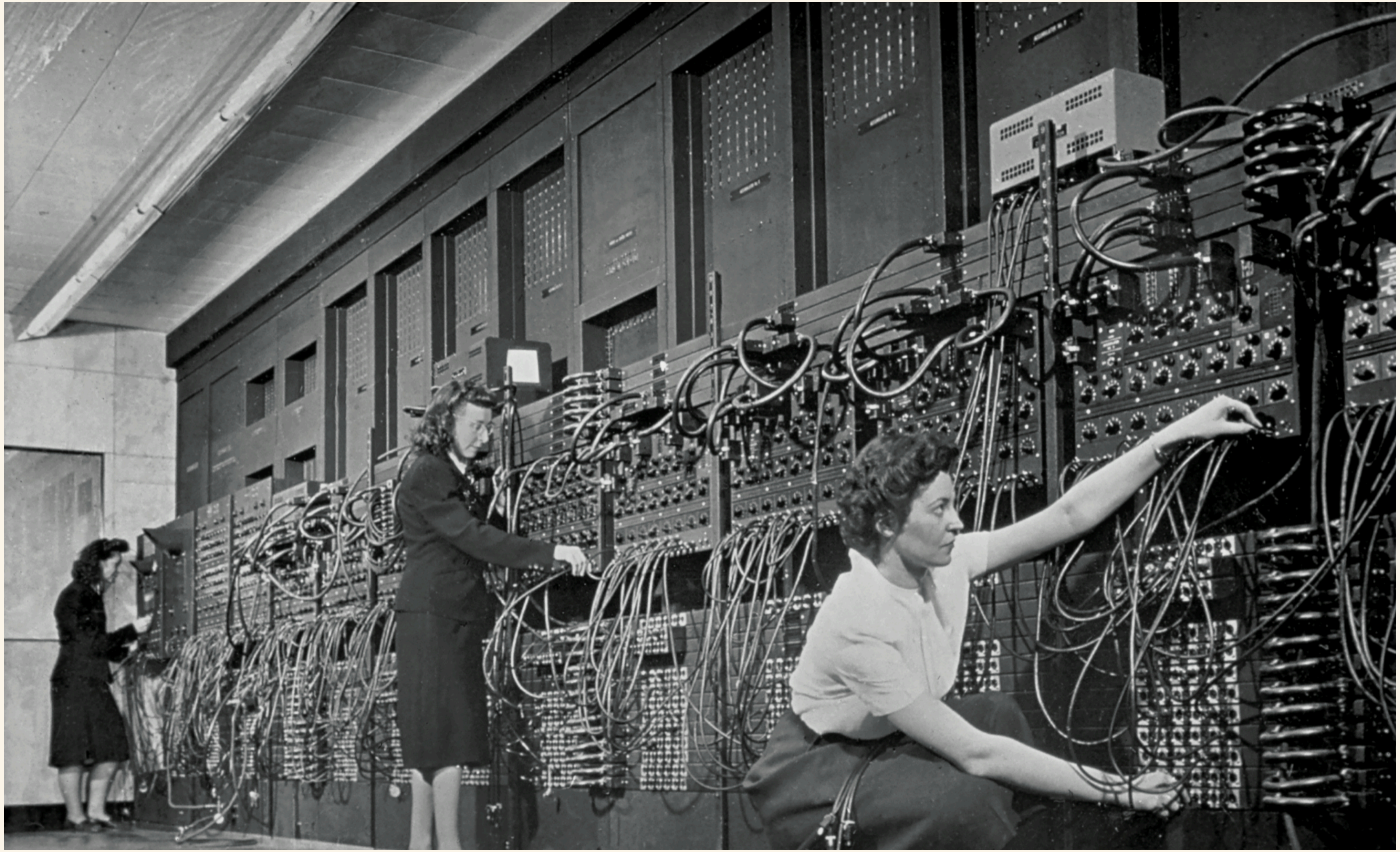


30 tons

18,000 vacuum tubes

Mean time between failures: hours

Programmed by rewiring



What was it for?

(no really — what would you have said in 1945?)

What they built it for

- Artillery firing tables
- Weather forecasting
- Cryptanalysis
- Census & payroll

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What it turned into

- Spreadsheets
- Social networks
- Search engines
- LLMs writing homework

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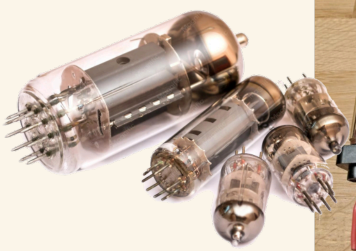
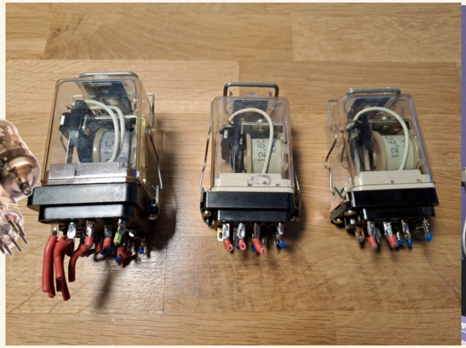
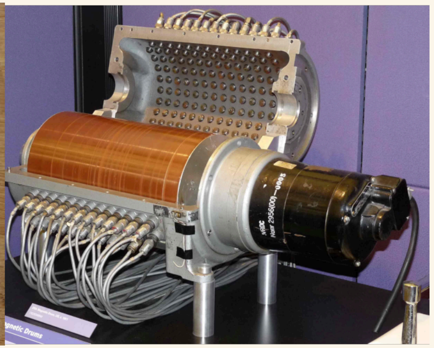
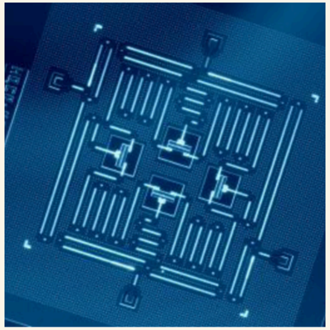
1945 $\rightarrow$ 2007 · ENIAC to iPhone · **62 years**
2026 $\rightarrow$ 2035 · Approx. time of CRQC · **9 years**

Six parallels

Hardware diversity before convergence

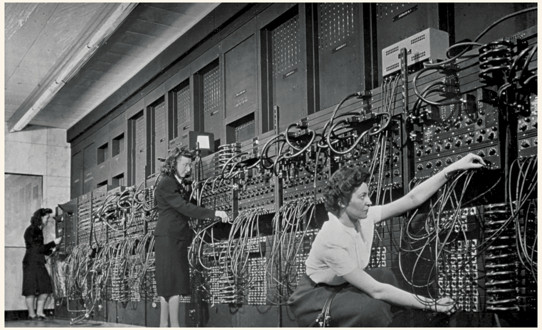
1945

Vacuum tubes
Relays
Mercury delay lines
Williams tubes
Magnetic drums
(all competing, none obvious)

Today

Superconducting (IBM, Google)
Trapped ions (IonQ, Quantinuum)
Neutral atoms (QuEra, Atom Computing)
Photonics (PsiQuantum, Xanadu)
Spin qubits (Intel), topological (MS)
(all competing, none obvious)

Specialist operators



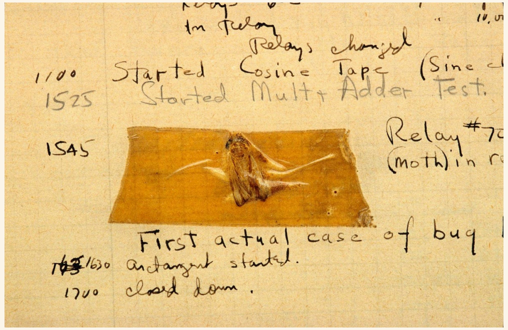
1945

The women who rewired ENIAC by hand
You don't walk up and use it
You knew the machine physically

Today

Cryogenic engineers
Calibration specialists
Pulse-level control experts
You submit a job to a queue

Error rates and reliability



1945

A vacuum tube fails every few hours
Uptime is a daily battle
Redundancy is the whole game

Today

Gate errors around 10^{-3}
Decoherence in microseconds
Error correction is the transistor moment

Programming model in flux

```
program hello
  ! This is a comment line;
  print *, 'Hello, World!'
end program hello
```

```
qreg q[4];
creg c[4];
h q[0];
cx q[0], q[1];
```

1945

1945: machine code, wired by hand
1949: assembly
1957: FORTRAN — the breakthrough abstraction

Today

Qiskit, Cirq, PennyLane — assembly stage
Braket, Q#, Quil — flavours of the same
The FORTRAN of quantum hasn't been written yet

Funding and access

1945	**Today**
Government, military, universities	US, EU, China, UK national initiatives
Manhattan Project money	IBM, Google, Microsoft, AWS
IBM, Bell Labs, MIT	A wave of startups

Killer-app uncertainty

We know quantum will be transformative for some problems.

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We don't yet know the full list.

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Factoring · Simulation of quantum systems · Certain optimisation · Certain ML

We had **Shor's algorithm (1994)** years before a machine that can run it.

The algorithm before the computer — surely ***that's*** *new?*

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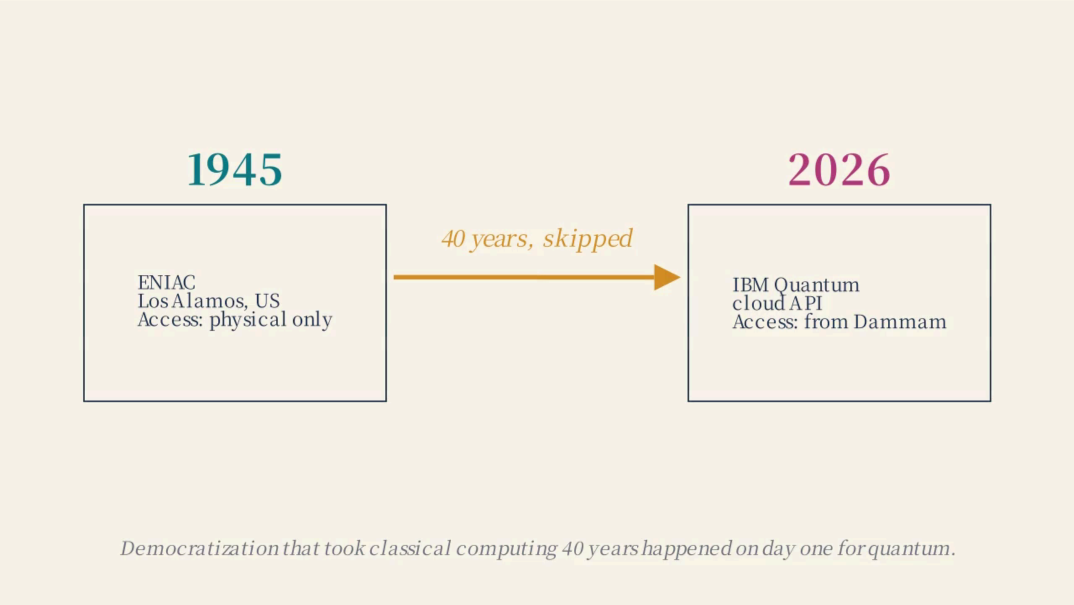
*The algorithm before the computer — surely **that's** new?*

Classical also was the same:

Ada Lovelace, 1843
The first algorithm — for Babbage's Analytical Engine, never built in her lifetime.

Alan Turing, 1936
Defined computation itself, nine years before ENIAC switched on.

But here's what's different


1945
ENIAC
Los Alamos, US
Access: physical only
40 years, skipped
2026
IBM Quantum
cloud API
Access: from Dammam
Democratization that took classical computing 40 years happened on day one for quantum.

So what actually is quantum?

Three experiments. No math.

Experiment 1

Double-slit, one photon at a time

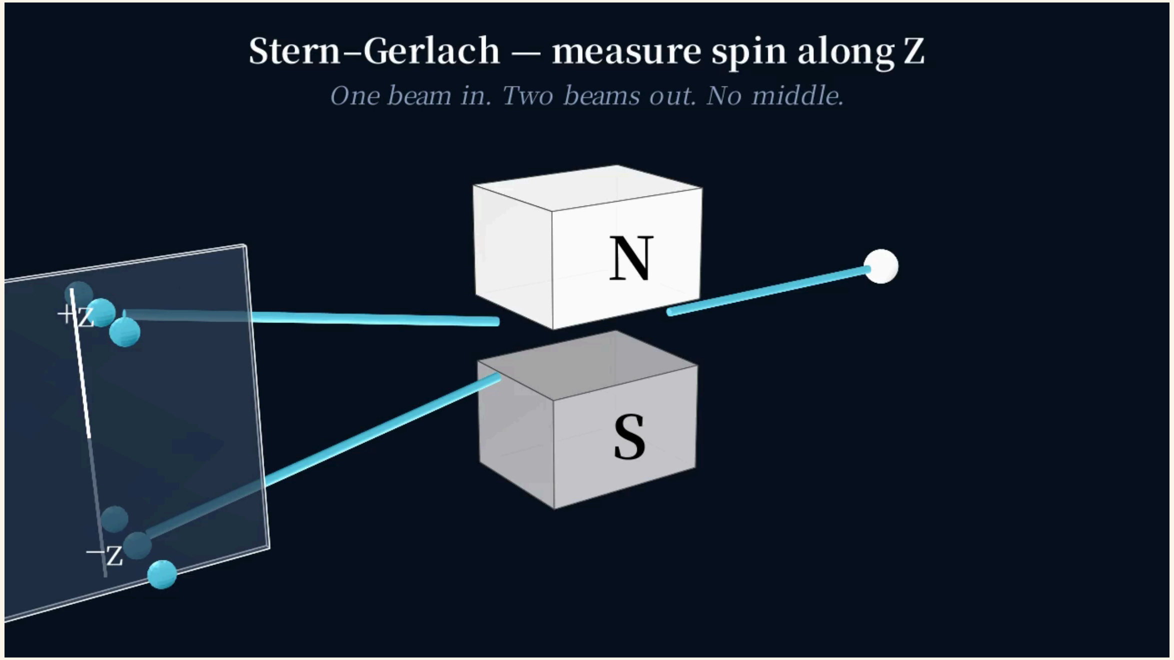
MANIM

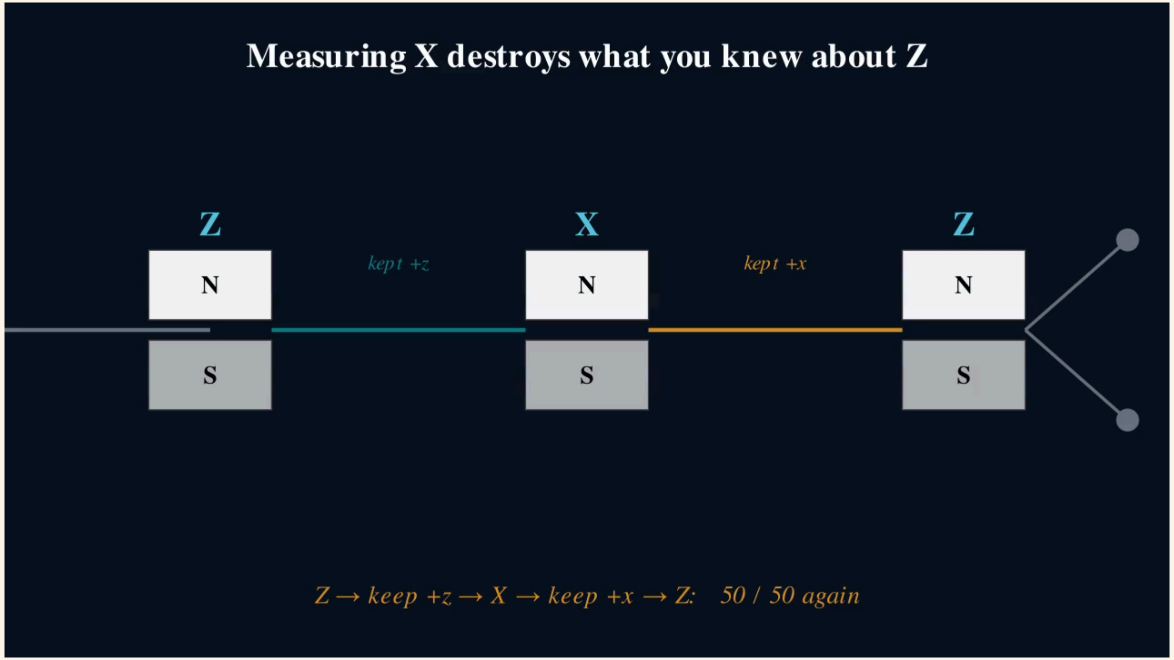
Hitachi single-electron buildup video · 2 min · side-channel MP4 · alt-tab to play

Particles interfere with themselves.

Experiment 2

Stern-Gerlach — measuring changes what you know


Stern–Gerlach — measure spin along Z
One beam in. Two beams out. No middle.
N
S
+Z
−Z


Measuring X destroys what you knew about Z
Z
X
Z
N
S
N
S
N
S
kept +z
kept +x
Z → keep +z → X → keep +x → Z: 50 / 50 again

Measuring one thing changes what you know about another.

Experiment 3

The CHSH game — beat the classical bound

Two volunteers, please.

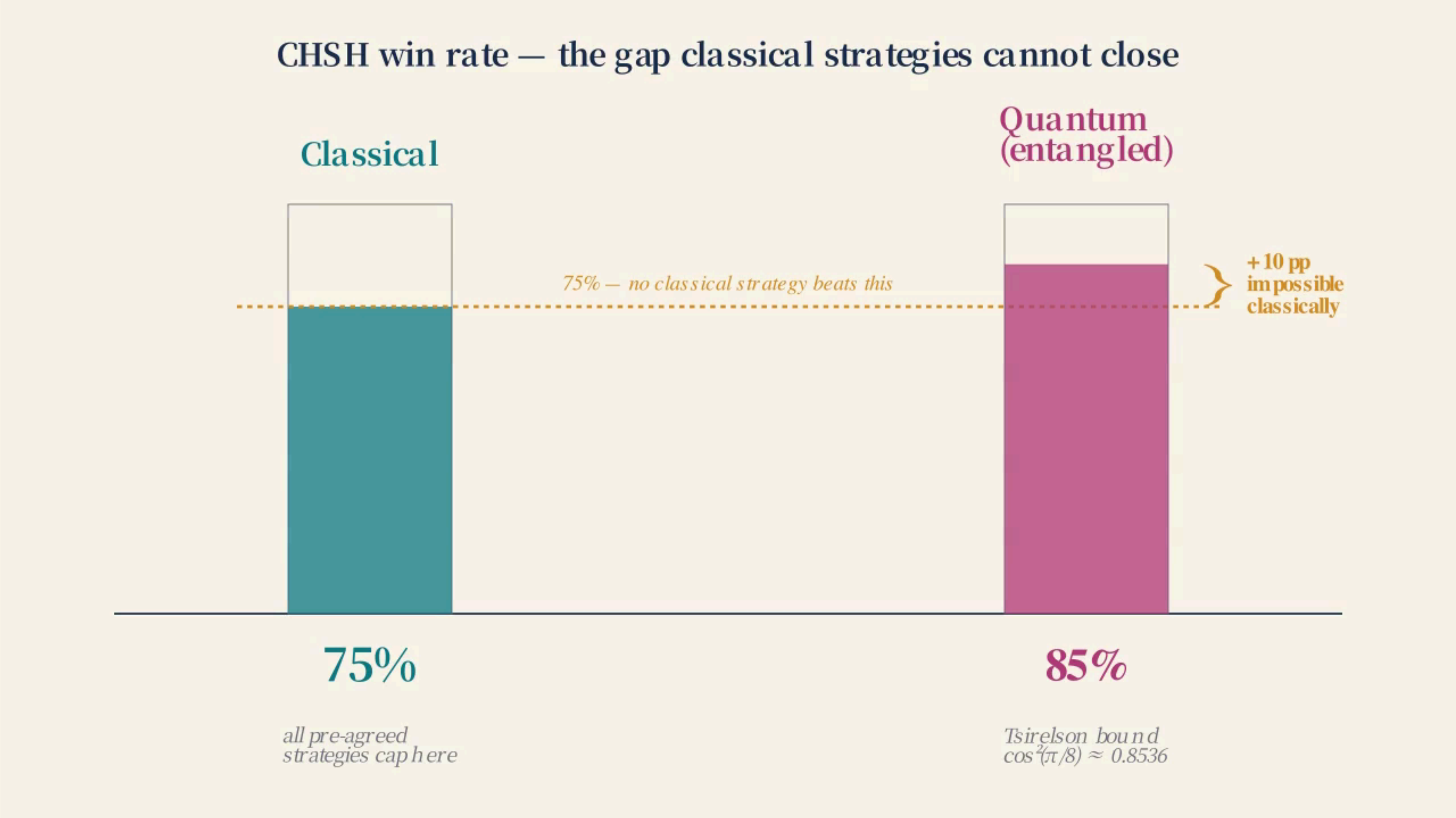
You'll play a game.

75% is the ceiling for anyone using classical physics

Alain Aspect, John Clauser, Anton Zeilinger

Nobel Prize in Physics, 2022

“For experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science”


CHSH win rate — the gap classical strategies cannot close
Classical
Quantum (entangled)
75% — no classical strategy beats this
+10 pp impossible classically
75%
85%
all pre-agreed strategies cap here
Tsirelson bound $\cos^2(\pi/8) \approx 0.8536$

The two qubits aren't two separate objects — they're one shared thing.
What Alice sees can't be described without mentioning Bob's setting.
The classical bound assumed independence. That assumption breaks.
This is what people mean by entanglement.

In 20 minutes you're going to entangle two qubits on a real superconducting chip in New York from your seat in Dammam.

The team that built ENIAC could not have imagined this.

After lunch

✓ Laptops out

✓ Wifi confirmed

✓ IBM Quantum account (or make one)

[QR code: quantum.ibm.com]